Band-1 Hadamard Order 8116 — Miyamoto (Paley-I Seed H_2028 q=2027 + C_2030) + Paley-II q=4057 IN SPK; Gap 2052 Still OPEN

H_2028 Paley-I PASS; C_2030 PASS; H_8116 Miyamoto PASS; H_8116 Paley-II q=4057 PASS → IN SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7988_Miyamoto`; `2026-09-16_Band1_Hadamard_7732_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_8116_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2028 and 8116 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I seed H_{2028} = Paley-I($q = 2027$) and symmetric Paley conference C_{2030} over GF(2029) ($q = 2029 \equiv 1 \pmod 4$), **and** classical Paley-II H_{8116} = Paley-II($q = 4057$) (prime $\equiv 1 \pmod 4$). Classical SPK for 8116: Paley-I **FAIL**s ($q = 8115 = 3 \cdot 5 \cdot 541$ composite); Paley-II **PASS**es ($q = 4057$ prime $\equiv 1 \pmod 4$); Kronecker **FAIL**s (4×2029). Therefore order **8116 is IN SPK via Paley-II** — Miyamoto alone never folds an order into SPK. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7988 Miyamoto-only NOT in SPK** (seed **1996** IN SPK). Prior SPK seeds **1996 / 1932 / 7604 / 1876+7508 / 1872 / 1860 / 7444 / 1848 / 7396 / 1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained; Miyamoto-only bonuses **7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **2028** and **8116** folded into ONBOX/SPK when verified (8116 via Paley-II only).

Abstract

Finite constructions of orders **2028** and **8116**.

- H_{2028} = Paley-I ($q = 2027$) — preferred seed (Paley-II $q = 1013$ alternate eligible unused; Double FAIL; Kronecker 4×507 FAIL)
- C_{2030} = Paley conference over GF(2029), $q = 2029$
- H_{8116} = Miyamoto(H_{2028}, C_{2030})
- H_{8116} = Paley-II ($q = 4057$) — classical SPK path
- Paley-I for 8116: **SKIP** ($q = 8115$ composite)
- Kronecker for 8116: **SKIP** ($4 \cdot 2029$ not admissible)

SPK: 8116 IN SPK via Paley-II $q = 4057$. Seeds 2028 and 8116 enter SPK. Prior SPK **1996 / 1932 / 7604 / 7508 / 1876 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8116.py`

Results. overall_pass=true; units 135/135; wall 878.3972s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 8116 = 4 \cdot 2029$; 2029 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{2028} and C_{2030} (Paley conference over prime field $q = 2029$).
- Seed: Paley-I $q = 2027$ (prime $\equiv 3 \pmod 4$) → H_{2028} (preferred); Paley-II $q = 1013$ alternate eligible; Double FAIL; Kronecker 4×507 FAIL.
- SPK: Paley-I $q = 8115$ **FAIL** (composite $3 \cdot 5 \cdot 541$); Paley-II $q = 4057$ **PASS** (prime $\equiv 1 \pmod 4$); Kronecker **FAIL → IN SPK via Paley-II**.
- Prior: 7988 Miyamoto-only shipped; this ship $q = 2029 \rightarrow n = 8116$ (+ Paley-II SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_2028 Paley-I	Paley-I q=2027 DIRECT	0	5.9655
C_2030	Paley conf GF(2029) q=2029	0	5.6119
H_8116 Miyamoto	Miyamoto(H_2028 Paley-I, C_2030)	0	419.3485
H_8116 Paley-II	Paley-II q=4057 (classical SPK)	0	437.1634
H_8116 Paley-I	SKIP q=8115=$3 \cdot 5 \cdot 541$	—	—
H_8116 Kronecker	SKIP $4 \cdot 2029$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **7988** lineage (**1996** Paley-II; **1932** Paley-I; **7604** Paley-I; **1876/7508**; **1872**; **1860/7444**; **1848/7396**; **1788**; **888/1776**; **860/1720/6884**; **1708**; **6724**; **840/1680**; **...**; **NOT** 7988/7732/7492/7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **2028** (Paley-I) + verified **8116** (Paley-II q=4057); **exclude** Miyamoto-only bonuses. Known count 1609. Next constructive gap **2052** (OPEN).

Quantity	Value
8116 in SPK	**True** (Paley-II q=4057)
2028 in SPK	**True** (Paley-I q=2027)
1996 / 1932 / 7604 / 7508 in SPK	**True** (prior)
7988 / 7732 / 7492 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · finite constructions of 2028+8116 ONLY · Miyamoto alone never folds into SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_8116_miyamoto/`
Artifacts: `H2028_paleyI_pm1.npz`, `C2030_pm1.npz`, `H8116_miyamoto_pm1.npz`, `H8116_paleyII_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.